

MAE SOT, Thailand

WMO: 483750

Lat: 16.7025N

Lon: 98.5417E

Elev: 197

StdP: 98.98

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	15.1	9.5	7.6	22.5	11.2	8.4	22.8	6.1	28.0	5.4	28.4	0.1	10	0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.0	37.2	24.2	36.2	23.8	35.2	23.7	26.7	32.1	26.2	31.5	25.9	31.1	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	28.8	25.0	20.5	28.3	24.7	20.1	27.9	85.0	32.4	83.1	31.8	81.6	31.3	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.3	4.5	DB	10.8	38.8	2.0	1.1	9.3	39.6	8.2	40.3	7.1	40.9	5.6	41.8	(4)
(5)			WB	10.0	28.4	1.9	1.7	8.6	29.6	7.5	30.7	6.4	31.6	5.0	32.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	23.8	25.7	27.9	30.0	29.0	27.4	26.5	26.5	27.2	27.3	25.9	23.9	(6)
(7)		DBStd	2.38	1.95	1.82	1.59	1.42	2.02	1.35	1.20	1.13	1.12	1.04	1.74	2.24	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD10.0	6117	429	439	556	601	591	523	512	511	516	536	476	429	(10)
(11)		CDD18.3	3077	171	206	298	351	332	273	253	252	266	278	226	172	(11)
(12)		CDH23.3	29698	1558	2212	3511	4453	3657	2428	1896	1845	2213	2475	1987	1463	(12)
(13)	CDH26.7	12159	623	1113	1926	2457	1634	703	423	401	646	902	792	539	(13)	
(14)	Wind	WSAvg	1.1	0.8	0.9	1.3	1.5	1.5	1.1	1.1	1.0	0.9	0.9	1.0	1.0	(14)
(15)	Precipitation	PrecAvg	1584	7	5	6	26	185	254	348	433	195	106	15	4	(15)
(16)		PrecMax	2452	41	23	27	83	381	450	914	822	391	260	60	15	(16)
(17)		PrecMin	1134	0	0	0	0	1	102	122	205	104	23	0	0	(17)
(18)		PrecStd	346	12	9	8	32	88	105	201	186	88	65	19	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	35.7	37.3	38.7	38.8	34.1	32.4	32.2	33.2	33.7	33.9	33.1	(19)
(20)			MCWB	22.1	21.8	22.8	24.4	25.1	26.0	25.8	25.7	25.9	25.3	24.6	23.1	(20)
(21)		2%	DB	32.3	34.6	36.3	37.6	36.8	32.6	31.2	31.1	32.1	32.7	32.9	32.0	(21)
(22)			MCWB	21.8	21.9	22.6	24.4	25.2	25.7	25.5	25.5	25.7	25.2	24.1	22.9	(22)
(23)		5%	DB	31.4	33.7	35.5	36.7	35.0	31.4	30.1	30.0	31.1	32.0	32.0	31.0	(23)
(24)			MCWB	21.5	21.7	22.6	24.4	25.3	25.4	25.1	25.3	25.5	25.1	23.8	22.4	(24)
(25)		10%	DB	30.3	32.7	34.5	35.7	33.2	30.2	29.0	28.9	30.1	31.1	31.0	29.9	(25)
(26)			MCWB	21.2	21.5	22.5	24.4	25.3	25.2	24.8	24.9	25.3	25.0	23.4	22.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	25.0	25.6	27.0	27.5	26.9	26.5	26.4	26.7	26.8	25.9	25.1	(27)
(28)			MCDWB	30.4	32.2	33.6	34.9	33.9	31.8	30.7	30.4	31.3	31.6	31.7	30.6	(28)
(29)		2%	WB	23.6	23.9	24.7	26.1	26.5	26.2	25.7	25.9	26.2	26.2	25.3	24.1	(29)
(30)			MCDWB	29.6	31.7	32.9	34.2	32.8	30.8	29.7	29.7	30.6	31.1	30.7	29.7	(30)
(31)		5%	WB	22.8	23.1	24.2	25.6	26.0	25.8	25.4	25.5	25.8	25.7	24.8	23.4	(31)
(32)			MCDWB	28.8	30.5	32.1	33.5	32.1	30.0	29.0	29.0	30.0	30.4	29.9	28.9	(32)
(33)		10%	WB	22.1	22.4	23.7	25.1	25.6	25.4	25.0	25.1	25.4	25.3	24.2	22.7	(33)
(34)			MCDWB	28.1	29.9	31.3	32.9	31.1	29.1	28.2	28.2	29.2	29.6	29.0	28.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.3	15.5	14.2	12.0	8.3	6.2	5.4	5.5	6.6	8.5	10.8	12.7	(35)
(36)			MCDBR	15.1	16.1	15.3	13.3	11.0	7.7	7.0	7.0	8.0	9.6	11.7	13.2	(36)
(37)			MCWBR	5.8	5.4	4.7	3.9	3.2	2.7	2.6	2.6	2.8	3.4	4.1	5.0	(37)
(38)		5% WB	MCDWB	13.0	13.9	13.1	11.8	9.0	6.9	6.2	6.2	7.4	8.6	10.4	11.9	(38)
(39)			MCWBR	5.6	5.2	4.4	3.8	3.1	2.8	2.7	2.6	2.9	3.3	3.9	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.448	0.531	0.651	0.603	0.494	0.444	0.435	0.438	0.456	0.521	0.455	0.445	(40)	
(41)		taud	2.177	1.969	1.695	1.868	2.215	2.383	2.410	2.390	2.324	2.087	2.243	2.213	(41)	
(42)		Ebn at Noon	835	781	703	740	814	847	856	861	848	781	823	825	(42)	
(43)		Edh at Noon	142	183	247	209	146	122	119	122	130	160	132	134	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.74	5.31	5.57	5.89	5.00	4.09	3.72	3.73	4.33	4.67	4.82	4.59	(44)	
(45)		RadStd	0.20	0.21	0.34	0.36	0.49	0.45	0.45	0.33	0.26	0.26	0.25	0.26	0.18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page